



Newton's Laws of Motion



CONNECTED PARTICLES



✓ Rest

$$T = mg$$

✓ $\uparrow$ $\overset{a=0}{\boxed{\text{const vel.}}} \downarrow$

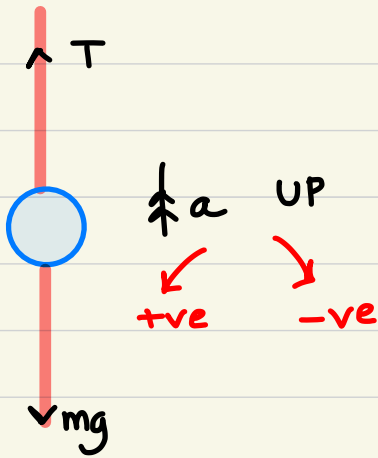
$$T = mg$$

UP $T - mg = m(0)$

DN $mg - T = m(0)$

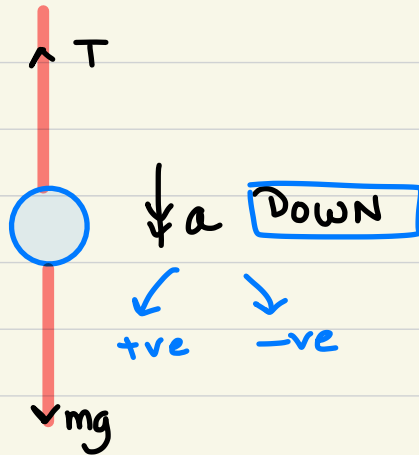
Accelerate
✓ $F = ma$

✓ Might is Right!



$$T > mg \text{ ignore}$$

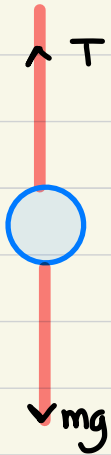
$$T - mg = ma \quad \checkmark$$



$$mg > T$$

$$mg - T = ma$$

$$F = ma, \text{ UP}$$



$\uparrow a$

$$m = 2 \text{ kg} \quad g = 10 \text{ m/s}^2$$

$$\text{Case I: UP} \quad 1 \text{ m/s}^2$$

$$T - mg = ma$$

$$T - 2(10) = 2(1)$$

$$T = 22 \text{ N}.$$

$$\text{Case: UP} \quad -1 \text{ m/s}^2$$

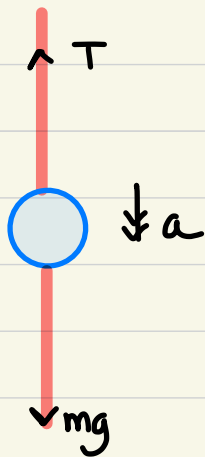
$$T - mg = ma$$

$$T - 2(10) = 2(-1)$$

$$T - 20 = -2$$

$$T = 18 \text{ N}$$

$$F = ma, \text{ Down}$$



$$\text{Case I : } a = 5 \text{ m/s}^2$$

$$mg - T = ma$$

$$2(10) - T = 2(5)$$

$$20 - T = 10 \quad \therefore T = 10 \text{ N.}$$

$$\text{Case II : } a = -5 \text{ m/s}^2$$

$$mg - T = ma$$

$$2(10) - T = 2(-5)$$

$$20 - T = -10$$

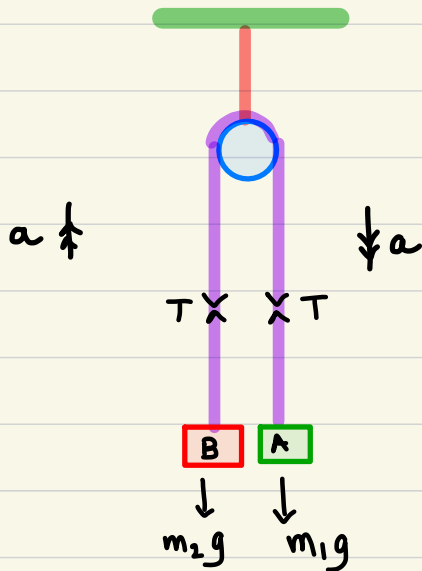
$$T = 30 \text{ N.}$$

CONNECTED PARTICLES

- ✓ particles
- ✓ system
- ✓ common acceleration $\Rightarrow$ accn. of the system.
- ✓ one string $\rightarrow$ one tension
Two strings $\rightarrow$ Two Tension
- ✓ Pulleys $\begin{cases} \rightarrow \text{smooth} \\ \rightarrow \text{fixed} \end{cases}$
- ✓ string $\begin{cases} \rightarrow \text{inextensible} \\ \rightarrow \text{Light} \end{cases}$

Pulley Systems

Case I : Simple Pulley



$m_1 = m_2 \rightarrow$ system remains at rest.

$$m_1 > m_2$$

particle A : $m_1 g - T = m_1 a$

particle B : $T - m_2 g = m_2 a$

system : $m_1 g - m_2 g = m_1 a + m_2 a$
 $= (m_1 + m_2) a$

$$m_1 = 0.6 \text{ kg} \quad m_2 = 0.4 \text{ kg} \quad g = 10 \text{ m/s}^2$$

$$0.6g - T = 0.6a$$

$$T - 0.4g = 0.4a$$

$$0.6g - 0.4g = (0.6 + 0.4)a$$

$$0.2g = 1.0a$$

$$0.2(10) = 1.0a$$

$$a = 2 \text{ m/s}^2$$

$$T - 0.4(10) = 0.4(2)$$

$$T - 4 = 0.8$$

$$T = 4.8 \text{ N}$$

$$m_1 = 0.6 \text{ kg}$$

$$m_2 = 0.4 \text{ kg}$$

$$g = 10 \text{ m/s}^2$$

Speed with which the heavier particle hits the floor.

$$u = 0 \text{ (Rest)}$$

$$a = 2 \text{ m/s}^2$$

$$s = 1 \text{ m}$$

$$v^2 = u^2 + 2as$$

$$= 0^2 + 2(2)(1)$$

$$v = 2 \text{ m/s} \checkmark$$

$$v = u + at$$

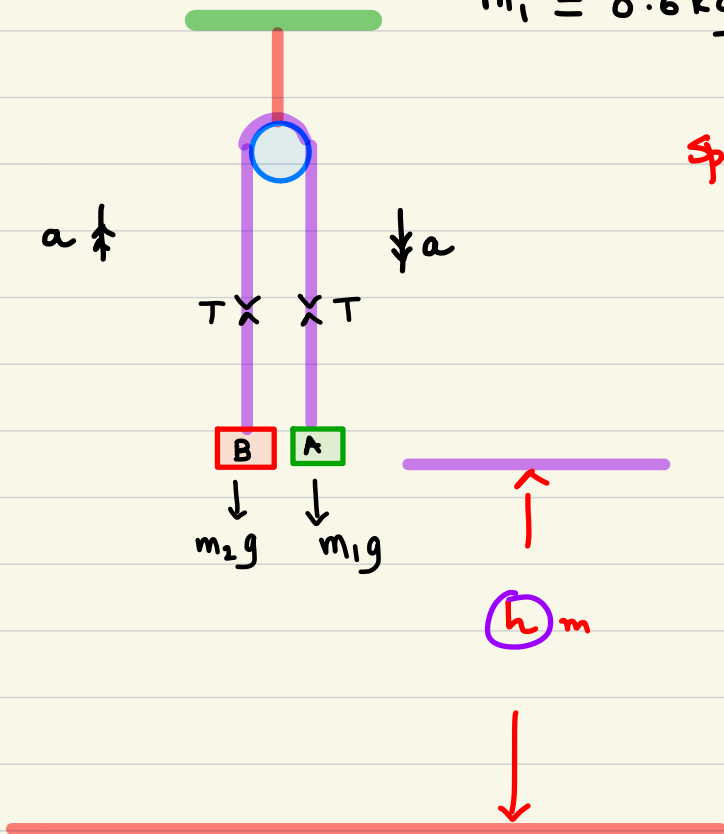
$$2 = 0 + 2(t) \quad \therefore t = 1 \text{ sec.}$$

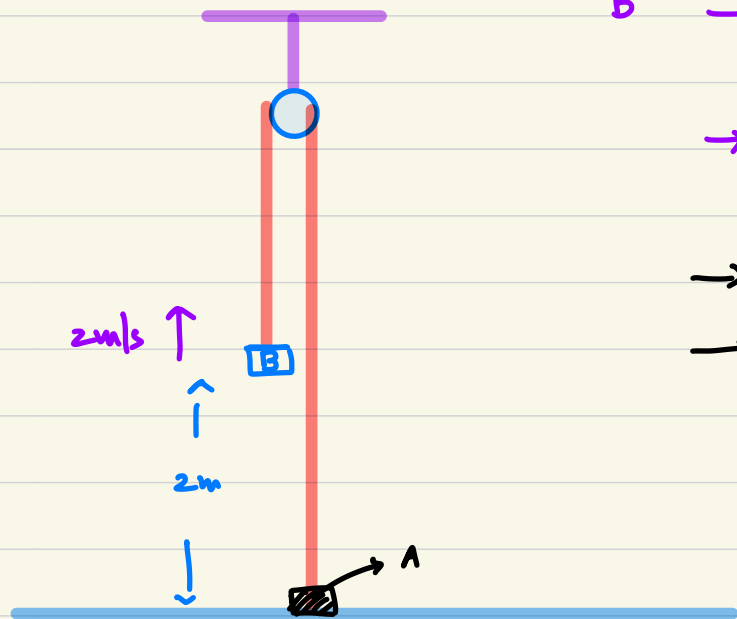
Max height above ground reached by the lighter particle (B).

particle A $\rightarrow$ hits the floor $\rightarrow 2 \text{ m/s}$

B $\rightarrow$ is Now 2m above the ground.

2m/s.





B → free particle

→ freely moving under gravity

→ string slack

→ No Tension in the string.

$$u = 2 \text{ m/s}$$

$$v = 0$$

$$a = -10 \text{ m/s}^2$$

$$v = u + at$$

$$0 = 2 + (-10)t$$

$$t = 0.2 \text{ sec}$$

$$v^2 = u^2 + 2as$$

$$0^2 = 2^2 + 2(-10)s$$

$$\frac{4}{20} = s \quad \text{OR} \quad s = 0.2 \text{ m}$$

Time to reach
max ht from
the moment it was released

$$\text{max ht} = 1 \text{ m} + 1 \text{ m} + 0.2 \text{ m} \\ \text{sys} \quad \text{free} \\ = 2.2 \text{ m}$$

$$1.0 \text{ sec} + 0.2 \text{ sec} \\ \text{system} \quad \text{free} \\ = 1.2 \text{ sec}$$

Taut

Lec #01

N. Laws